Diagnoses, Admission Type, and Hospital Readmission

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CAI5735: Applied Data Science in Health | Group 8

Research Question

How is the number of diagnoses related to hospital readmission, and does that relationship differ by admission type (Emergency, Urgent, Elective)?

Columns: number_diagnoses, readmitted, admission_type

Dataset

3,000 hospital stays of patients with diabetes, from 130 US hospitals (course sample, no missing values)

Source: UCI Machine Learning Repository; Strack et al. (2014), doi:10.24432/C5230J

number_diagnoses: diagnoses recorded for the stay

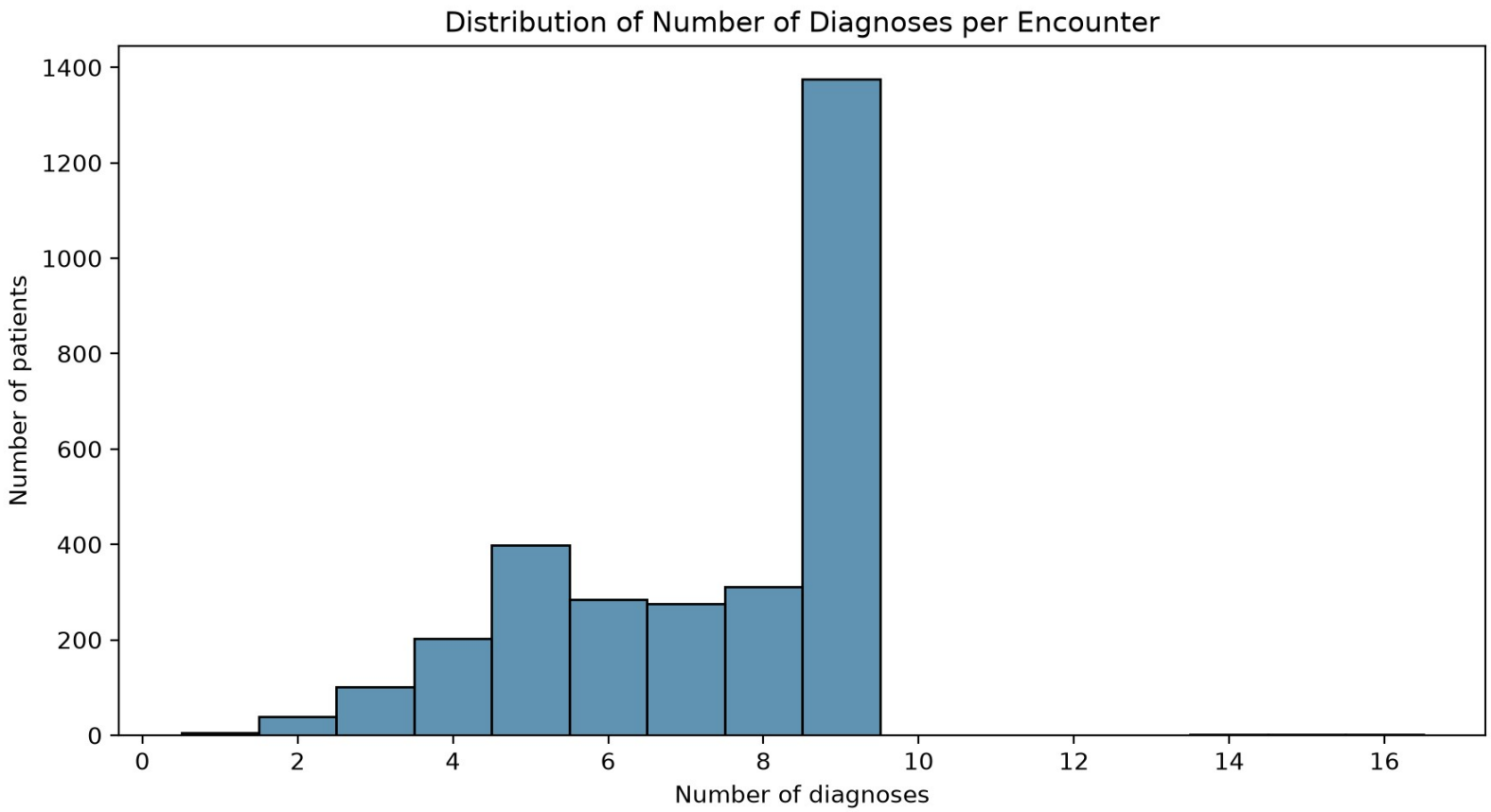
readmitted: not readmitted (60%), after 30 days (31%), within 30 days (9%)

admission_type: Emergency (57%), Elective (22%), Urgent (21%)

Analysis Process

1. Kept the three relevant columns
2. Added computed columns: readmitted flag (0/1) and diagnosis group (1-4, 5-6, 7-8, 9+)
3. Filtered to Emergency admissions as a check within one type
4. Grouped by admission type and diagnosis group; sorted by readmission rate
5. Made a histogram, boxplot, scatter plot, and group comparison charts

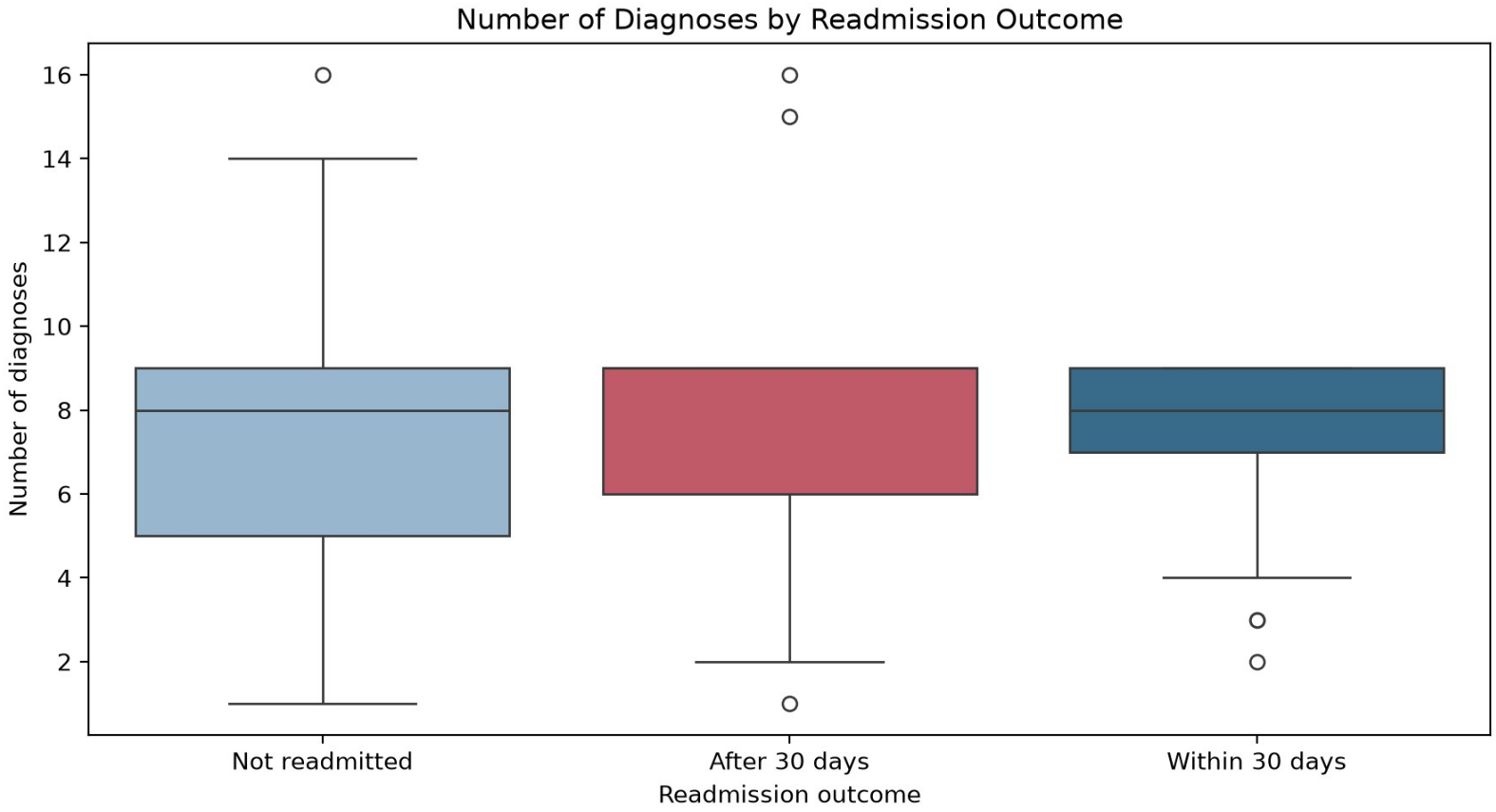
Most stays have 9 diagnoses



1,376 of 3,000 stays have exactly 9 diagnoses

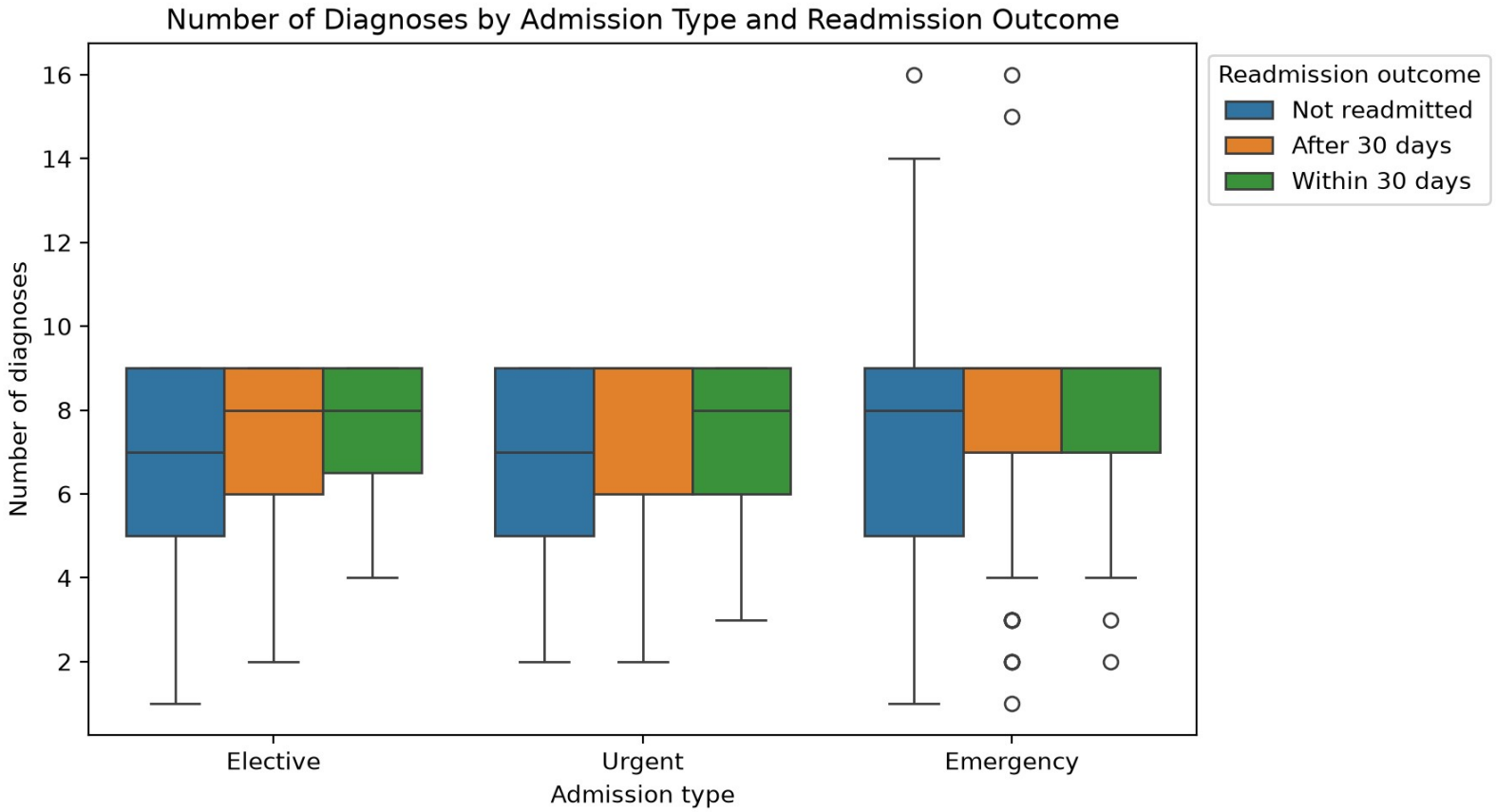
Likely because the source hospital systems capped recorded diagnoses, so patients with more conditions may be undercounted

Readmitted patients have more diagnoses



Average diagnoses: 7.0 for not readmitted vs 7.7 for readmitted (either timing)

Readmitted patients: more diagnoses in all types



Average diagnoses, not readmitted vs readmitted (either timing):

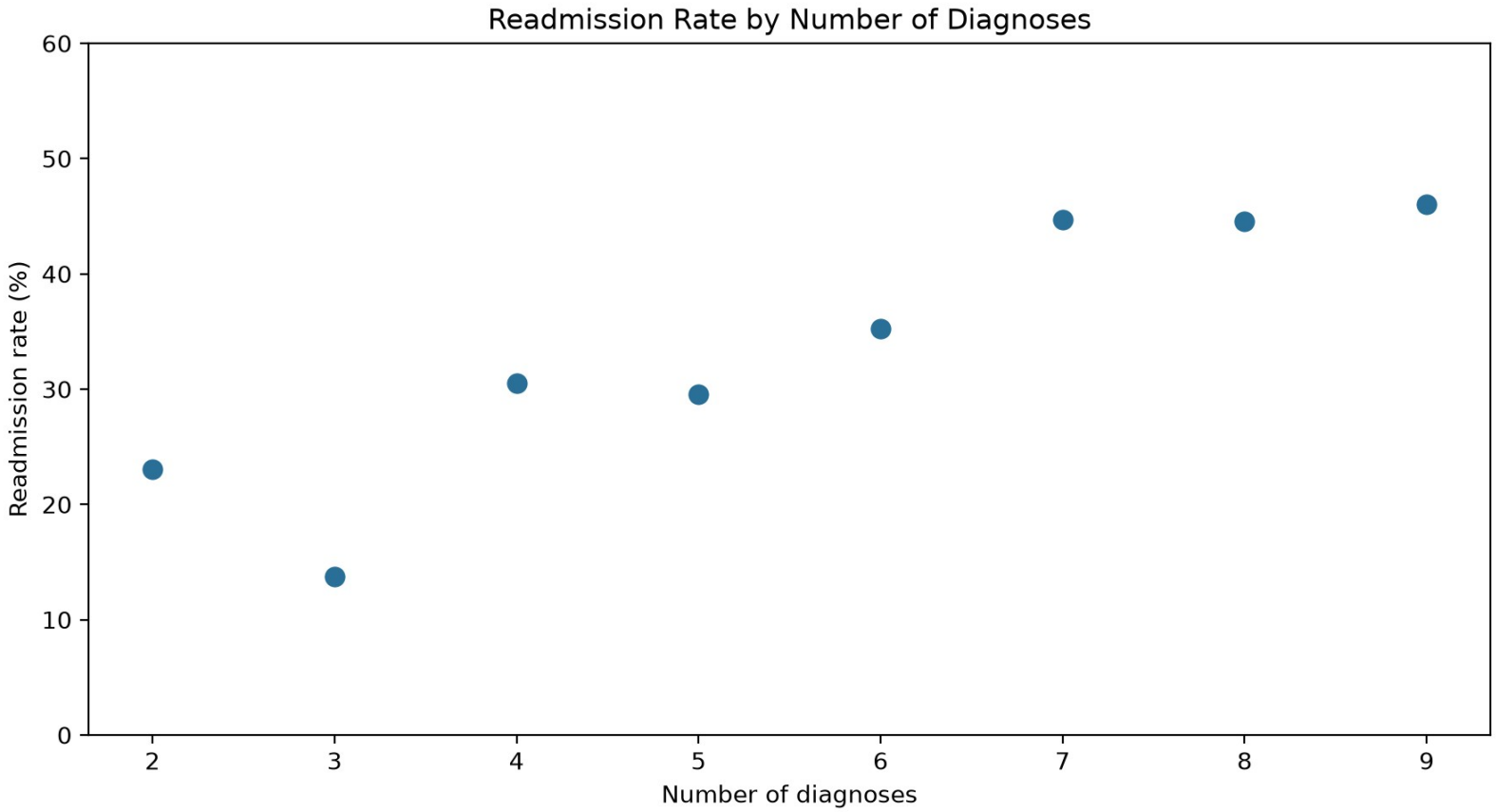
Elective: 6.7 vs 7.4-7.7

Urgent: 6.8 vs 7.4-7.7

Emergency: 7.2 vs 7.7

The gap is smallest for Emergency

Readmission rises with diagnoses

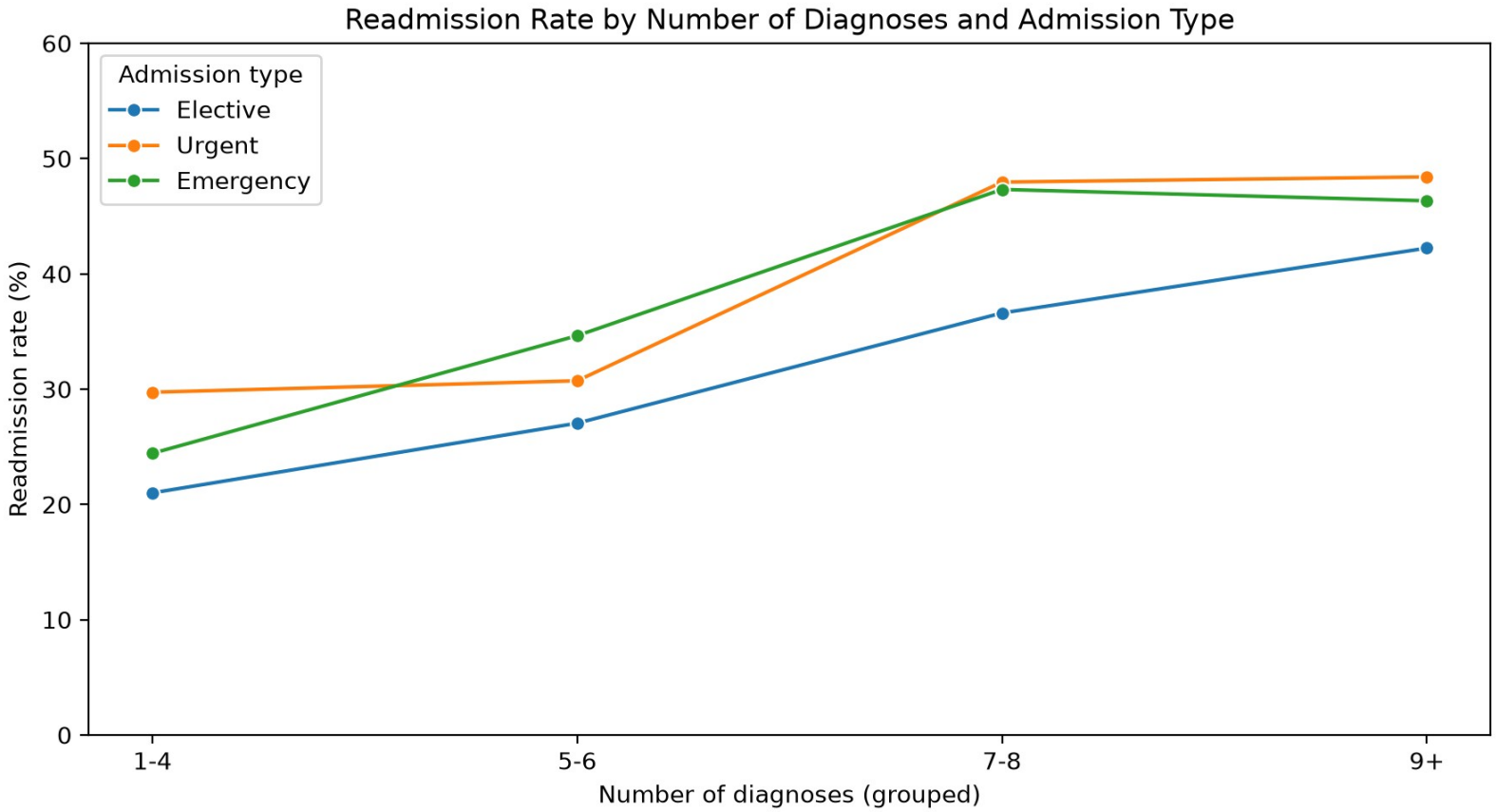


Rate climbs from 30% at 5 diagnoses to 46% at 9

Points at 2-3 diagnoses are noisy (39 and 102 patients)

Overall: 40.0% readmitted

Same upward trend in each admission type



1-4 diagnoses: 24.6% readmitted; 9+: 46.0%

Elective is lowest in every group (34.1% vs 41.4% Urgent, 41.8% Emergency)

Emergency-only check shows the same rise

Key Takeaways

More diagnoses goes with higher readmission, leveling off after about 7

Readmission rises in all three admission types; Elective is consistently lowest

Limits: association, not causation; many stays have exactly 9 diagnoses; small subgroups (Urgent 1-4 has 74 patients)

Questions?

github.com/apriljwalter-lgtm/Group_8_Midterm_project